Historical Air Quality Analysis

Air Quality UCI dataset · Italian city · March 2004 – April 2005

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Executive Summary

This report analyses 9,357 hourly observations of five gaseous pollutants (CO, NMHC, C_6H_6, NOx, NO_2) and three meteorological variables (temperature, relative humidity, absolute humidity) collected by a multi-sensor field device in an Italian city between 10 March 2004 and 04 April 2005. The objective is to identify temporal patterns, quantify pollutant relationships, and assess what these patterns imply for public health and urban policy.

Headline findings:

- Traffic dominates the air-quality signal. CO, benzene and NOx all show a sharp twin-peak diurnal cycle with maxima at 08:00 and 19:00 — the morning and evening commutes — and a deep trough at 05:00. Mean CO at 19:00 is roughly five times the 05:00 mean.
- **Benzene is consistently above WHO reference levels.** The annual mean of 10.1 µg/m³ is several times the WHO long-term benchmark of about 1.7 µg/m³ (excess-cancer-risk-of-1×10^{-5}), indicating a chronic public-health concern over the sampled year.
- **Pollutants are highly co-correlated.** CO and benzene correlate at r = 0.91, and CO–NOx at r = 0.80, consistent with a common combustion source (road traffic).
- **Weekend NOx falls ≈ 27 % below weekday levels,** a textbook signature of commuter and freight traffic. Weather variables on their own are weak predictors of pollution but interact strongly with NOx and NO_2.
- **A simple Random-Forest model predicts hourly benzene with $R^2 \approx 0.77$** from CO, NOx, NO_2 and weather alone, demonstrating that low-cost proxy sensors plus weather data can usefully estimate benzene without a dedicated benzene analyser.

1. Dataset and Methodology

1.1 Source

Data are taken from the UCI Machine Learning Repository "Air Quality" dataset (De Vito et al., University of Salerno). Hourly averages were recorded by a multi-sensor device deployed at road level in a significantly polluted area of an Italian city.

1.2 Variables

Variable	Units	Description
CO(GT)	mg/m³	Reference CO analyser (true value)
C6H6(GT)	µg/m³	Reference benzene analyser
NOx(GT)	ppb	Reference NOx analyser
NO2(GT)	µg/m³	Reference NO_2 analyser
PT08.S1–S5	—	Tin-oxide sensor responses (CO, NMHC, NOx, NO_2, O_3)
T, RH, AH	°C, %, —	Air temperature, relative humidity, absolute humidity

1.3 Cleaning steps

- Replaced the −200 sentinel value with NaN throughout.
- Combined Date and Time columns into a Python datetime index.
- Dropped NMHC(GT) — over 90 % missing across the period.
- Time-interpolated short gaps (≤ 6 hours); longer gaps left as NaN.
- Removed two empty trailing columns and 114 fully-blank trailing rows.

2. Temporal Trends

2.1 Daily and seasonal patterns

Figure 1 plots the daily mean of each pollutant overlaid with a 7-day rolling mean. CO and benzene show a mild seasonal dip in late summer (lower heating demand, longer photochemical lifetimes for some VOCs, holiday season). NOx and NO_2 rise sharply from October onwards — a familiar autumn/winter pattern driven by reduced atmospheric mixing (shallow boundary layer) plus increased space heating. The dataset ends before the 2005 summer trough is visible.

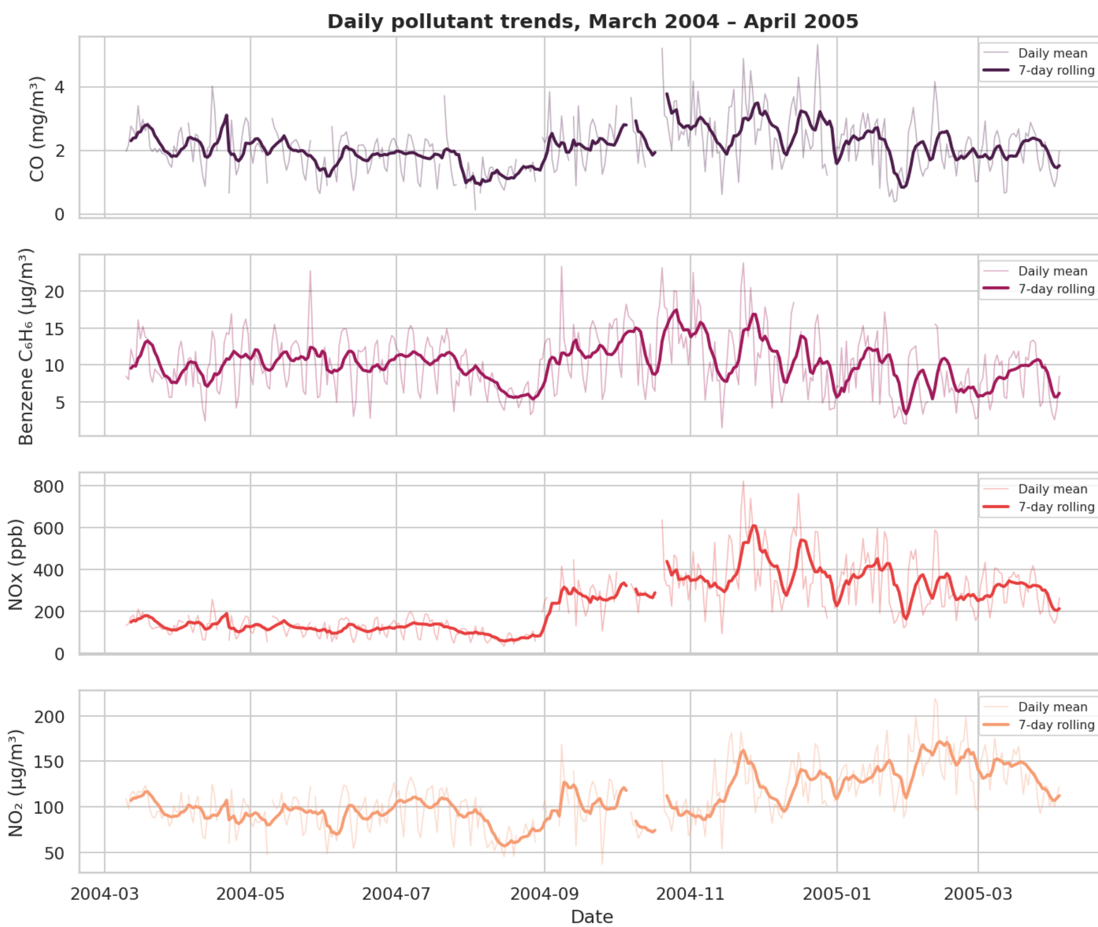

Figure 1 — Daily-mean concentrations (thin) with 7-day rolling mean (thick).

2.2 Monthly view

Normalising each pollutant to its own peak makes the seasonal structure easier to compare across species (Figure 2). All four reach near-maximum values in November–February; CO and benzene reach a clear minimum in August.

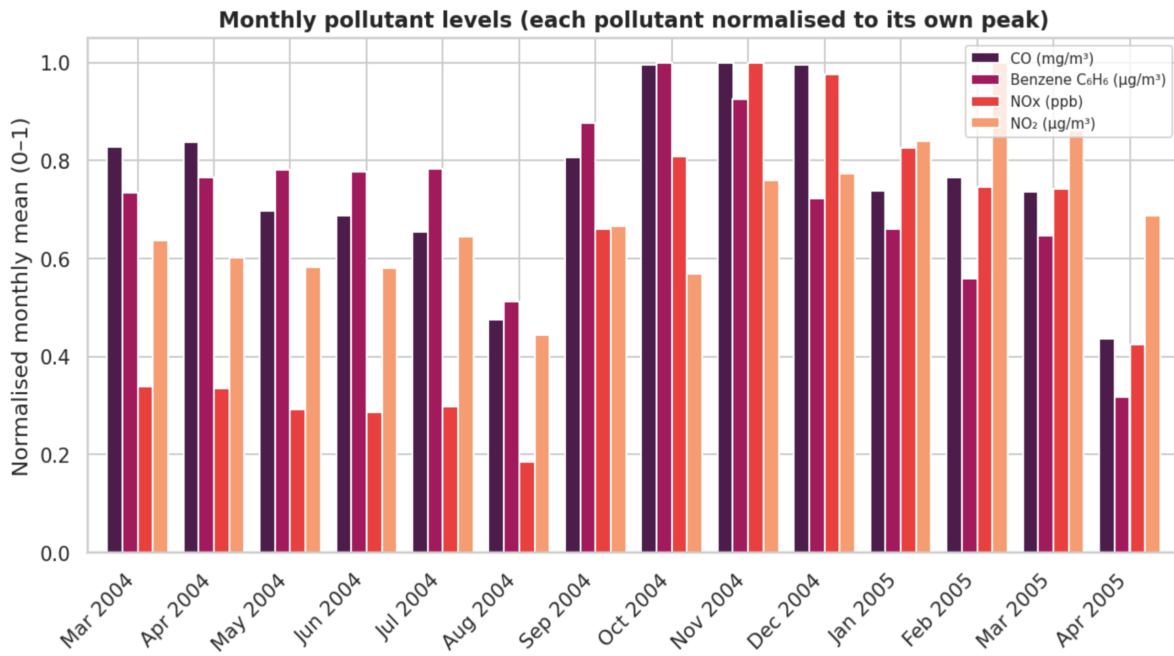

Figure 2 — Monthly means, each pollutant rescaled to its own maximum.

2.3 Diurnal cycle

Figure 3 collapses the time series onto the 24-hour clock. The twin commute peaks at 08:00 and 19:00 are unmistakable, with the evening peak the larger of the two. NO_2 stays elevated through the afternoon — consistent with photochemical conversion of NO emitted in the morning peak.

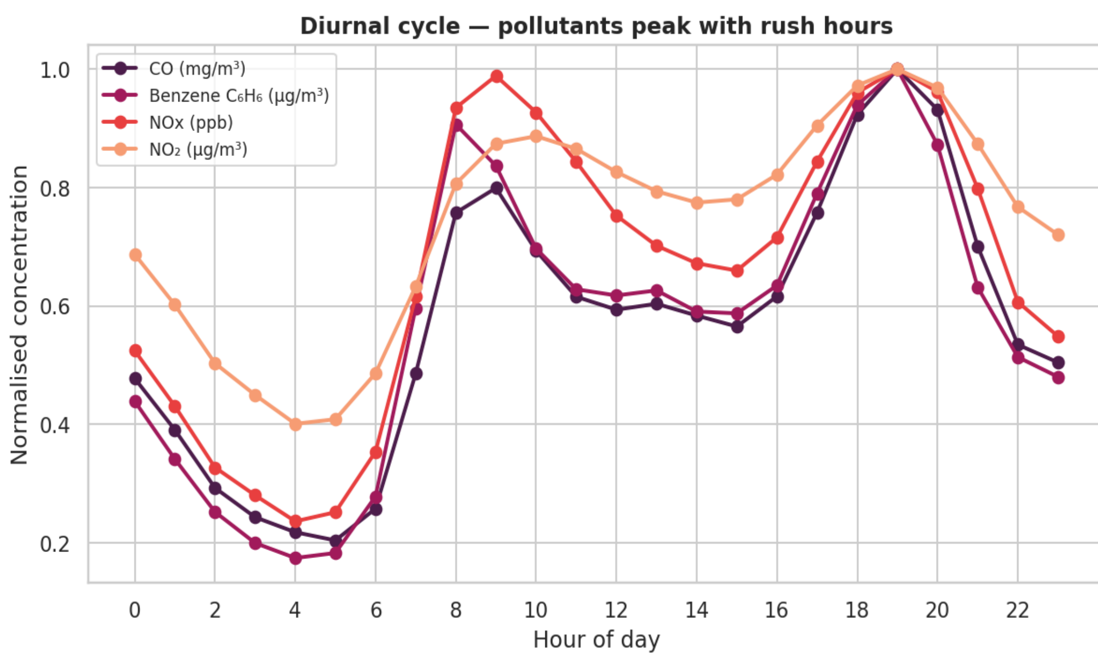

Figure 3 — Mean diurnal profile, normalised within each pollutant.

2.4 Weekday vs weekend

Box plots in Figure 4 show distributions splitting Monday–Friday from Saturday–Sunday. Median NOx falls by about 27 % at weekends and CO by about 18 %. NO_2 shows a smaller weekend reduction, reflecting its longer atmospheric lifetime and partial formation from background NO.

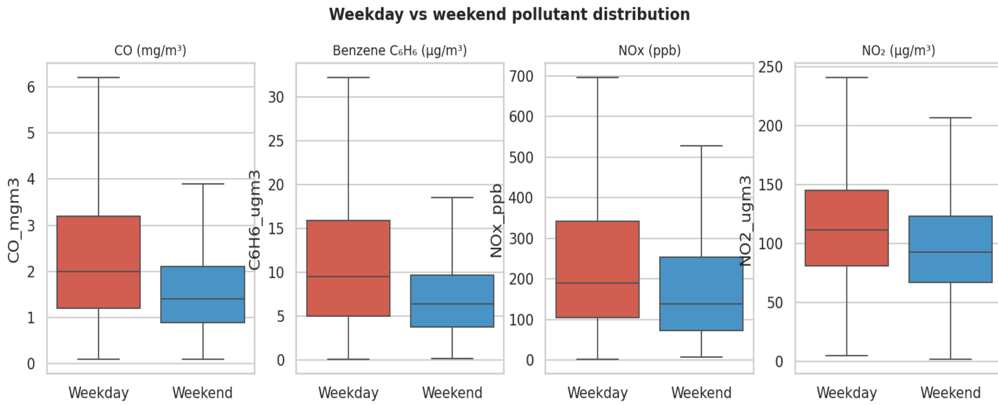

Figure 4 — Weekday vs weekend distributions (whiskers at 1.5×IQR; outliers hidden).

3. Pollutant–Weather Relationships

3.1 Correlation matrix

Figure 5 shows pairwise Pearson correlations. Three blocks stand out:

- **Combustion cluster:** CO, benzene, NOx and NO_2 are mutually correlated ($r \approx 0.6 - 0.9$), pointing to road traffic as the dominant common source.
- **Weather cluster:** Temperature and absolute humidity correlate strongly and positively ($r = 0.66$) because warm air holds more water vapour. Temperature and relative humidity correlate negatively ($r = -0.58$), the standard inverse relationship.
- **Cross terms:** NOx and NO_2 correlate weakly but negatively with temperature ($r \approx -0.2$), reflecting better dispersion in warmer, more convective conditions; NO_2 correlates negatively with absolute humidity ($r = -0.33$).

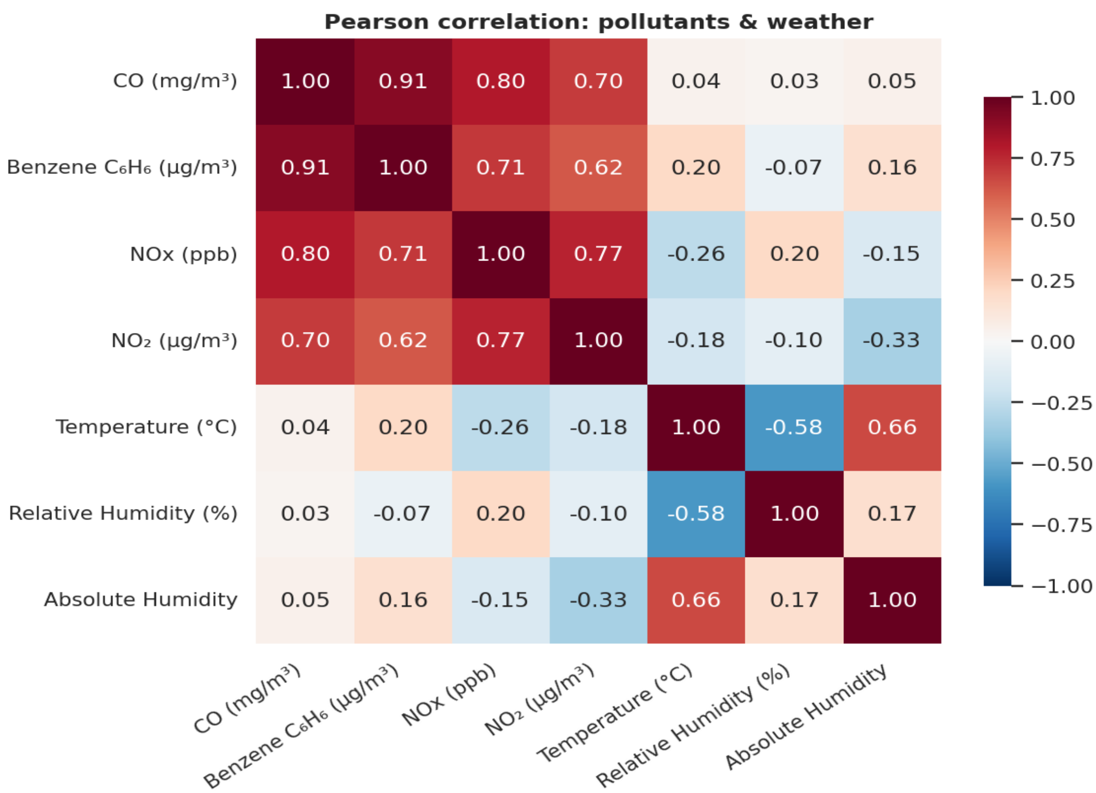

Figure 5 — Pearson correlation matrix for the four reference pollutants and three weather variables.

3.2 Scatter views

Figure 6 illustrates the cross-pollutant–weather relationships. The CO–temperature relationship is essentially flat ($r \approx 0.04$): CO depends much more on traffic intensity than on weather. NO_2 shows a modest negative slope against relative humidity, although a substantial amount of variance remains unexplained.

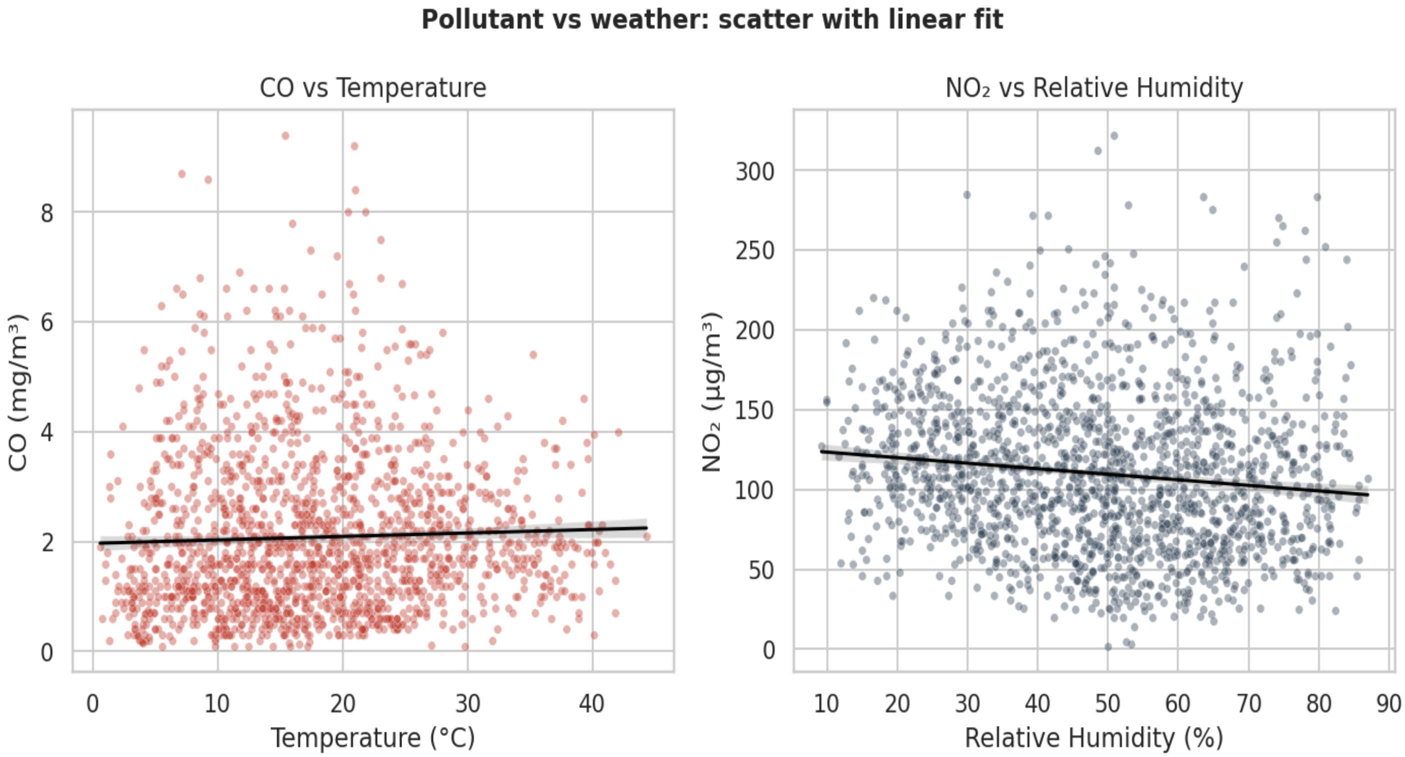

Figure 6 — Scatter plots with linear fit (random sample of 2,000 hours).

4. Predictive Modelling

As an optional analytical step, two regression models were trained to predict hourly benzene (C_6H_6) from co-located readings of CO, NOx, NO_2, weather, and time-of-day features. The dataset was split chronologically — first 80 % for training, final 20 % for testing — to avoid temporal leakage.

Model	MAE (µg/m³)	R^2
Linear regression	2.26	0.77
Random forest (120 trees)	1.76	0.77

Both models reach $R^2 \approx 0.77$; the Random Forest cuts mean absolute error by about 22 % thanks to its ability to capture non-linear interactions. Predicted vs actual benzene over the held-out period (Figure 7) tracks the diurnal and weekly cycles closely; the model underestimates the highest spikes, a typical regression-to-the-mean behaviour.

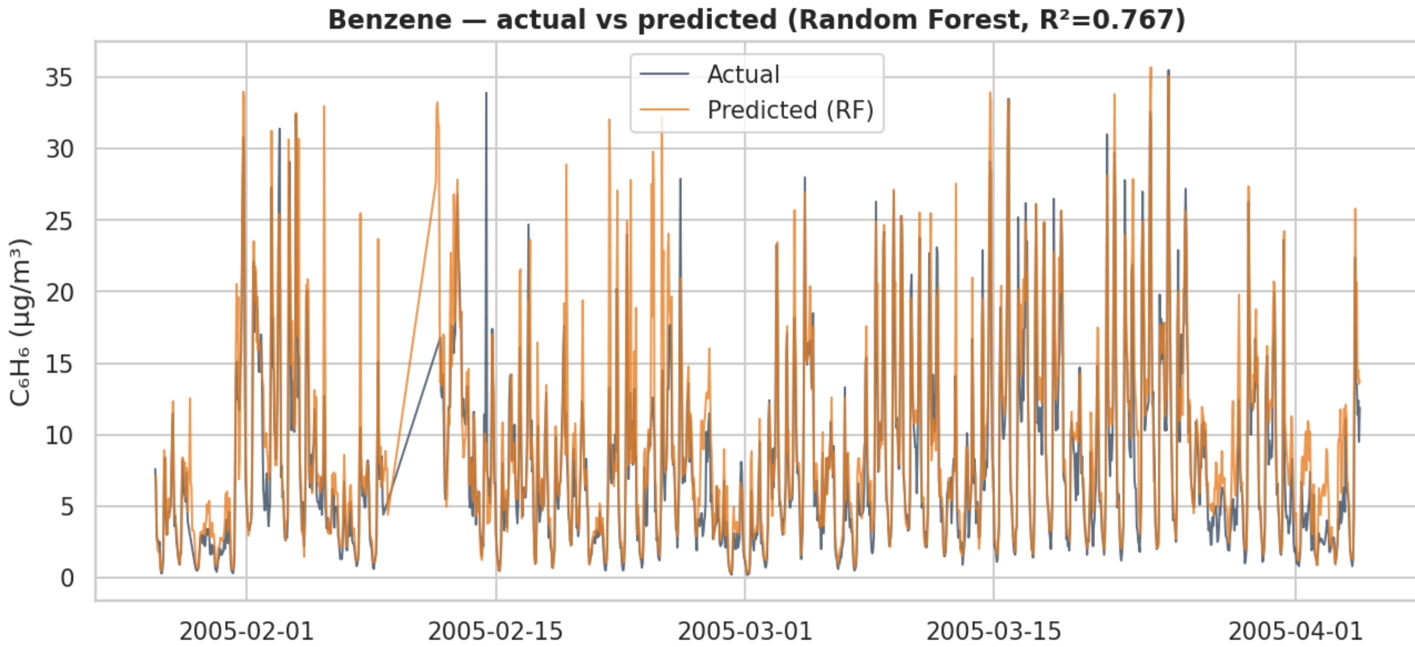

Figure 7 — Random-Forest benzene predictions on the held-out 20 % of the timeline.

Feature importance (Figure 8) is dominated by CO — unsurprising given the r = 0.91 correlation between CO and benzene noted earlier, since both are emitted in proportion to incomplete combustion of petrol-like fuels. NOx and time-of-day features make moderate contributions; weather variables are individually weak predictors.

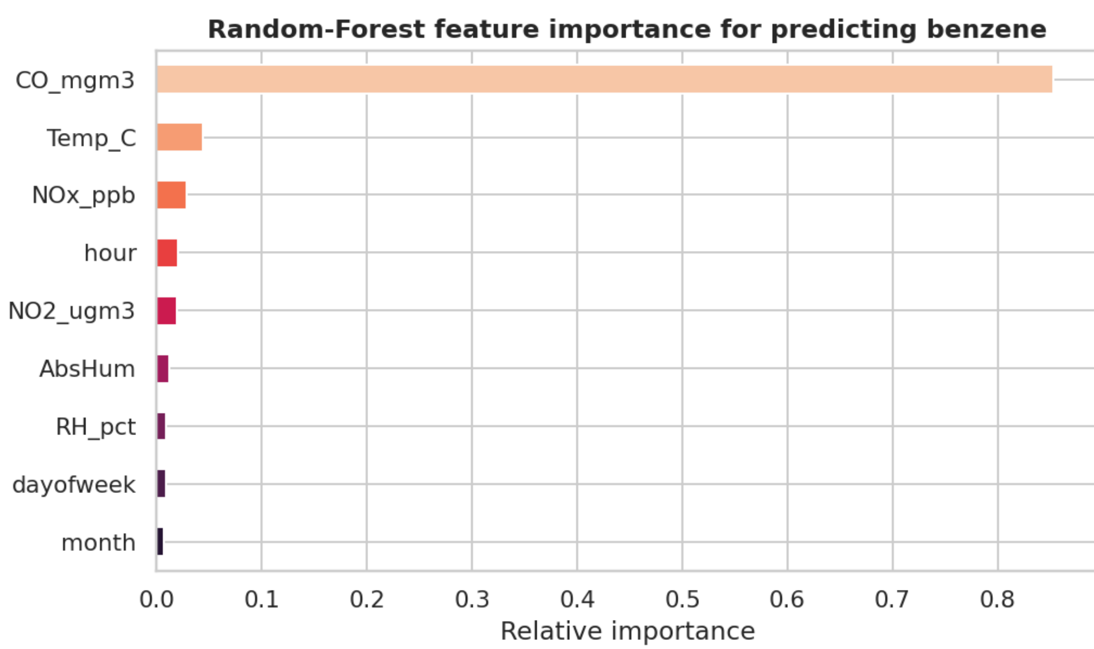

Figure 8 — Random-Forest feature importance for the benzene model.

5. Discussion and Implications

5.1 Public-health implications

Benzene is a Group-1 human carcinogen (IARC). The annual-mean concentration in this dataset (10.1 µg/m³) is roughly six times the WHO reference level associated with a 1-in-100 000 lifetime excess-cancer risk, and twice the EU statutory annual limit value (5 µg/m³, Directive 2008/50/EC). Sustained exposure at these levels warrants policy attention.

NO_2 is associated with respiratory inflammation, asthma exacerbation and reduced lung function in children. Hourly readings frequently exceed 200 µg/m³ during evening commutes, which is the EU short-term limit value (1-hour mean not to be exceeded more than 18 times per year).

5.2 Policy implications

- The 27 % weekend NOx drop demonstrates that traffic-management measures (low-emission zones, congestion charging, work-from-home incentives) can deliver measurable air-quality gains.
- Winter peaks suggest a co-benefit case for cleaner residential heating and a shift away from solid-fuel heating in colder months.
- The strong CO–benzene correlation indicates that benzene control is achievable largely through the same vehicle-emission interventions that target CO and NOx.

5.3 Limitations

- Single-site dataset: results are not directly transferable to other urban contexts.
- 13-month coverage prevents proper inter-annual trend or policy-impact assessment.
- NMHC was discarded due to >90 % missingness, leaving total non-methane hydrocarbons unmeasured.
- Correlation does not equal causation; weather effects are partially confounded with seasonal traffic patterns.

5.4 Recommended next steps

- Compare against contemporaneous regulatory monitoring stations to validate sensor accuracy.
- Extend the model to ozone (O_3) using the PT08.S5 sensor, which was omitted here.
- Apply a state-space or seasonal ARIMA approach for formal trend extraction.
- Investigate hourly exceedances of EU short-term limit values for NO_2 and consider an alerting system.

Appendix · Summary Statistics

Variable	n	Mean	Std	Min	Median	Max
CO (mg/m³)	8,176	2.13	1.46	0.10	1.80	11.90
C_6H_6 (µg/m³)	9,117	10.14	7.51	0.10	8.30	63.70
NOx (ppb)	8,295	240.85	209.99	2	175	1479
NO_2 (µg/m³)	8,293	111.03	48.28	2	107	340
Temp (°C)	9,117	18.31	8.83	-1.9	17.8	44.6
RH (%)	9,117	49.17	17.30	9.2	49.5	88.7